PROJECT PRESENTATION

LearnMate

AI assistant for smarter learning and study support

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Backend: AI StudyBuddy • Node.js • Express.js • MongoDB • Google Gemini API

The Problem: Study Overload

Meet Rahul

Second-year engineering student preparing for semester exams while learning skills for internships and placements.

Notes, PDFs, textbooks, articles and handwritten pages are scattered everywhere.

1. Lengthy material, limited time
2. Manual note-making takes hours
3. Flashcards and questions are repetitive
4. No personalised study plan
5. No secure place for AI resources
6. Hard to self-assess with quizzes

Solution & Key Features

1 Secure Login

JWT authentication, bcryptjs-hashed passwords

2 Material Upload

Upload notes as file or text, stored in MongoDB

3 AI Summaries

Concise summaries via Google Gemini

4 AI Flashcards

Question–answer cards for active recall

5 AI Quizzes

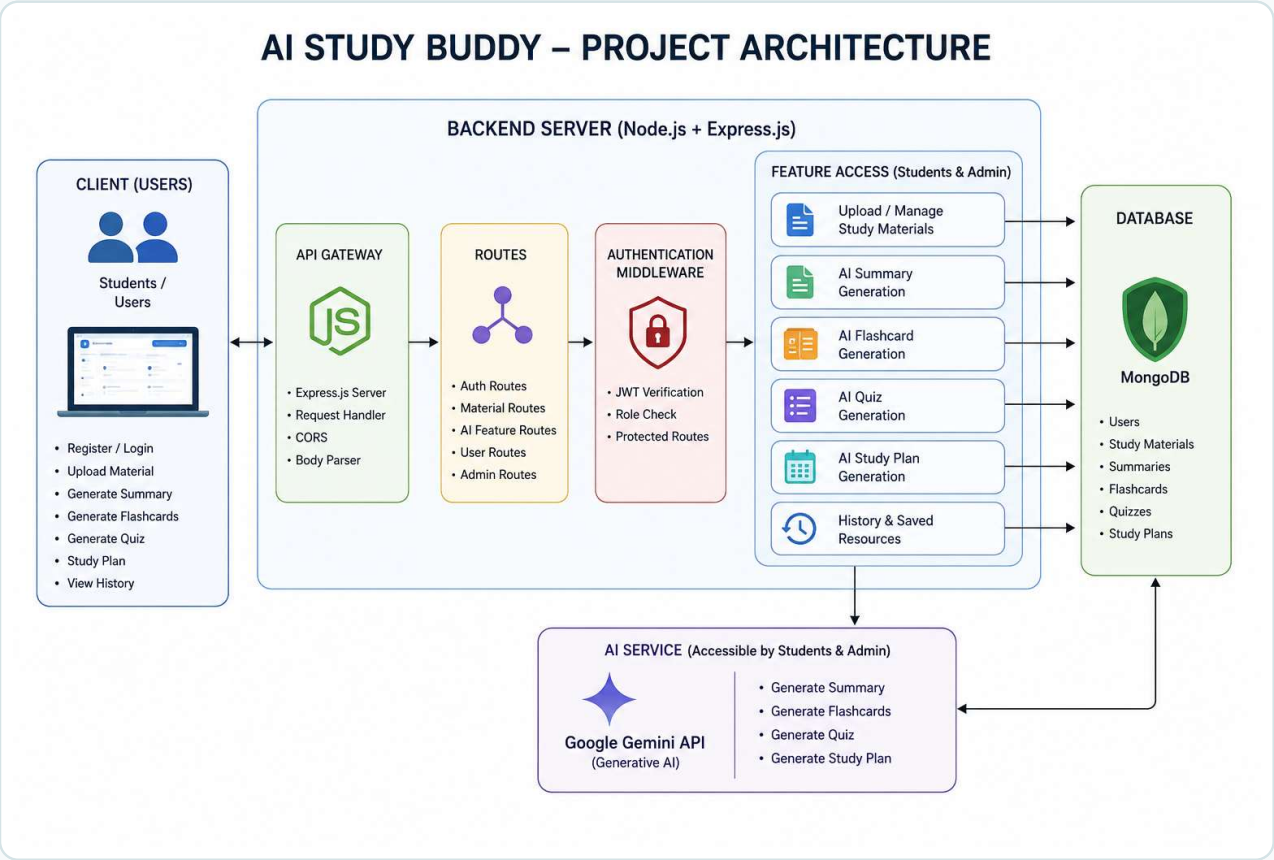
Multiple-choice self-assessment

6 AI Study Plans

Schedules from goals, time and exam date

Role-Based Access Control separates Student and Administrator privileges.

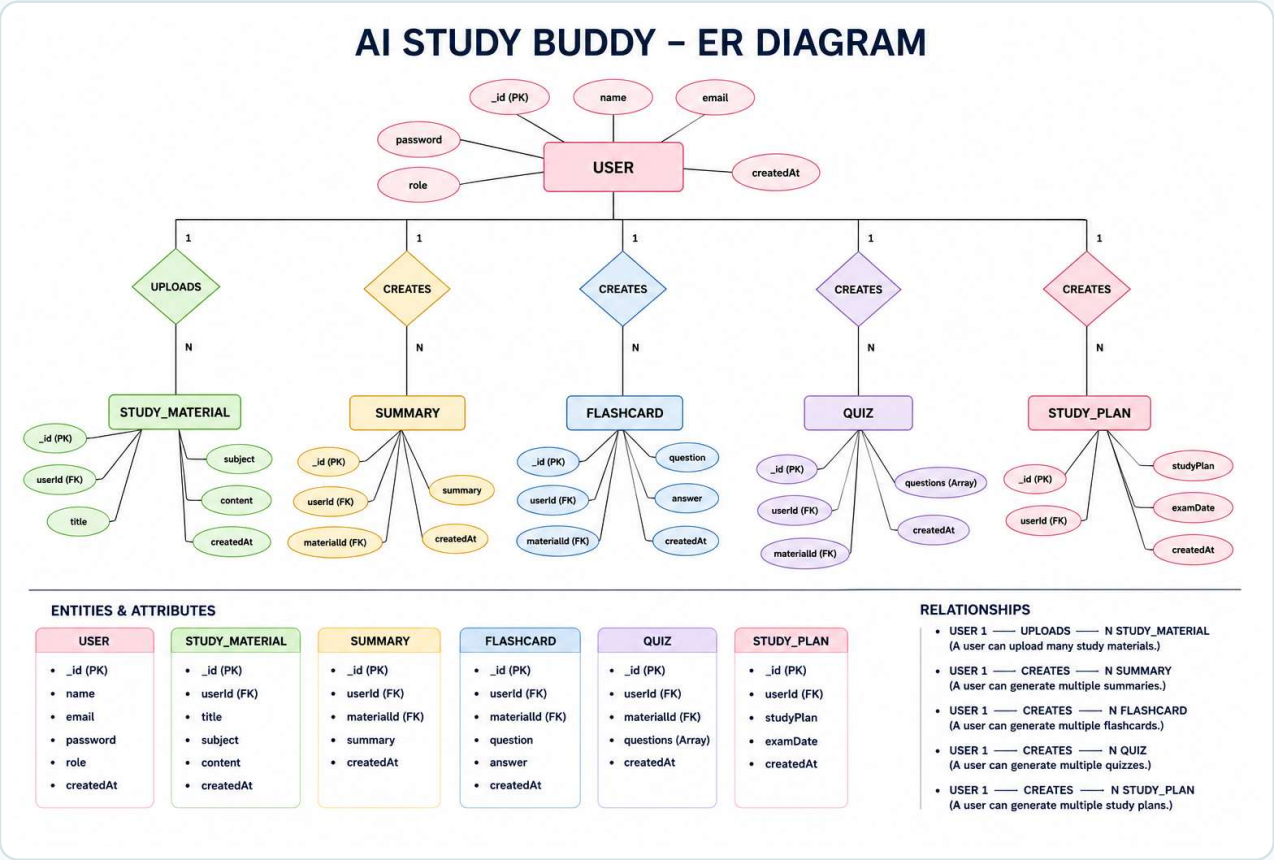
System Architecture



Modular REST design

- React client sends requests to the Express server.
- JWT middleware guards private routes.
- AI service layer calls Google Gemini.
- Mongoose stores users, materials and AI output in MongoDB.

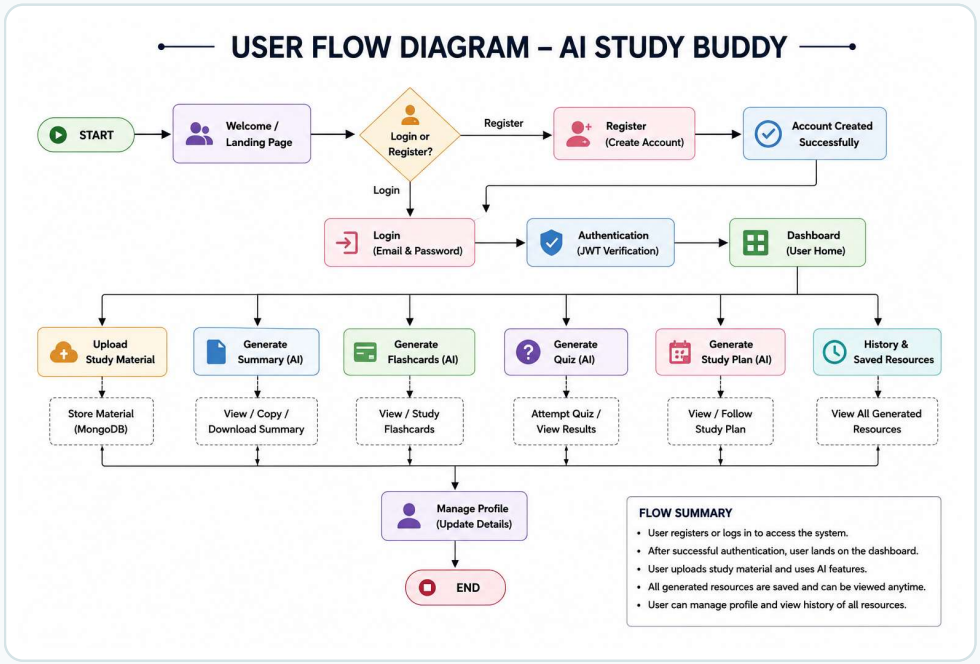
Data Design: ER Diagram



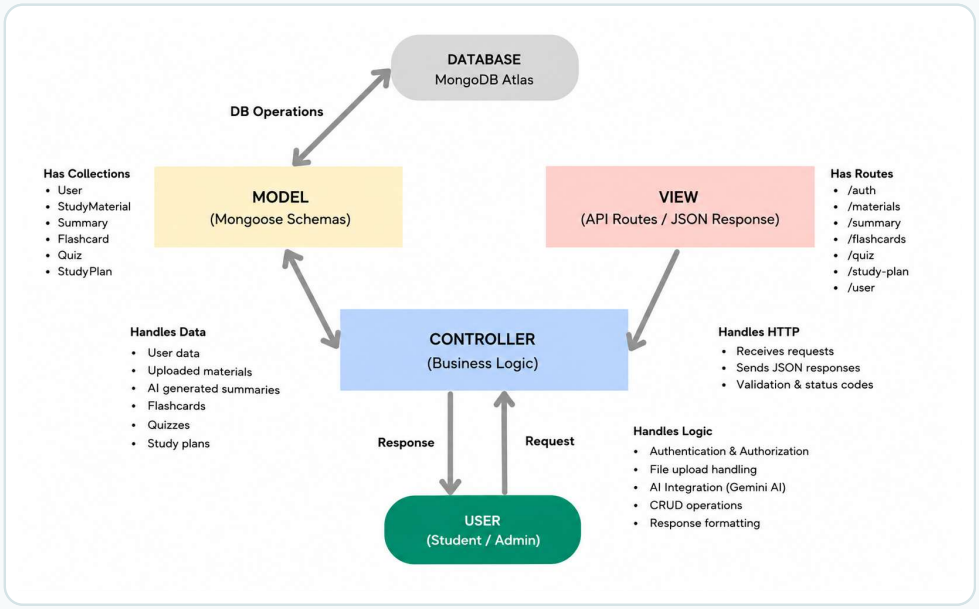
Key relationships

- User → Materials: one-to-many
- Material → Summary: one-to-one
- Material → Flashcards: one-to-many
- Material → Quiz: one-to-one
- User → Study Plans: one-to-many

User Flow & MVC Pattern



User flow: login → dashboard → upload → AI feature → save



MVC: Model (schemas), View (JSON), Controller (logic)

Separating data, logic and responses makes the code easier to test, debug and extend.

Development

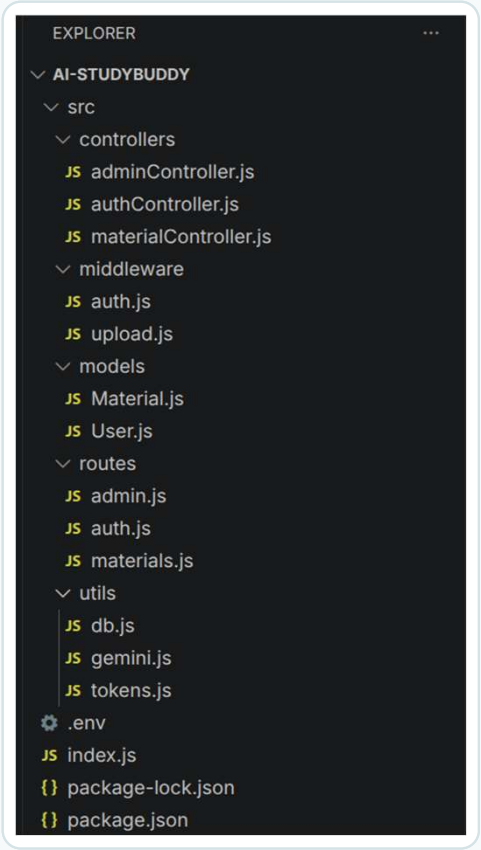
Node.js + Express.js
REST API server

MongoDB + Mongoose
Data and schemas

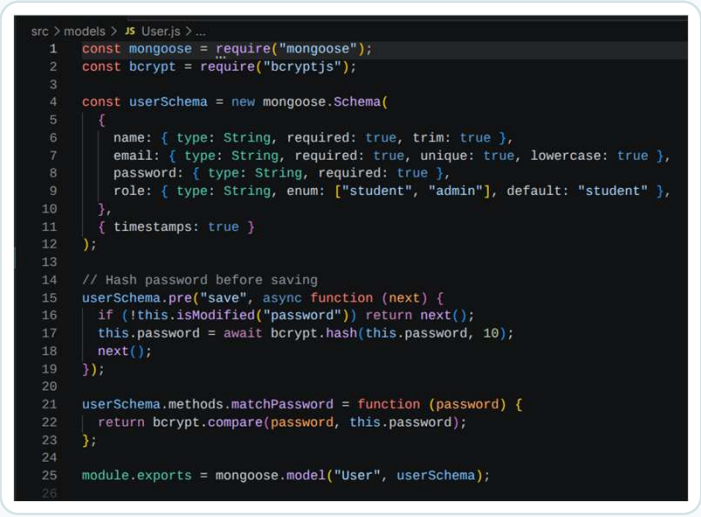
JWT + bcryptjs
Secure access

Gemini 2.5 Flash
AI generation

Multer
File uploads



Backend structure

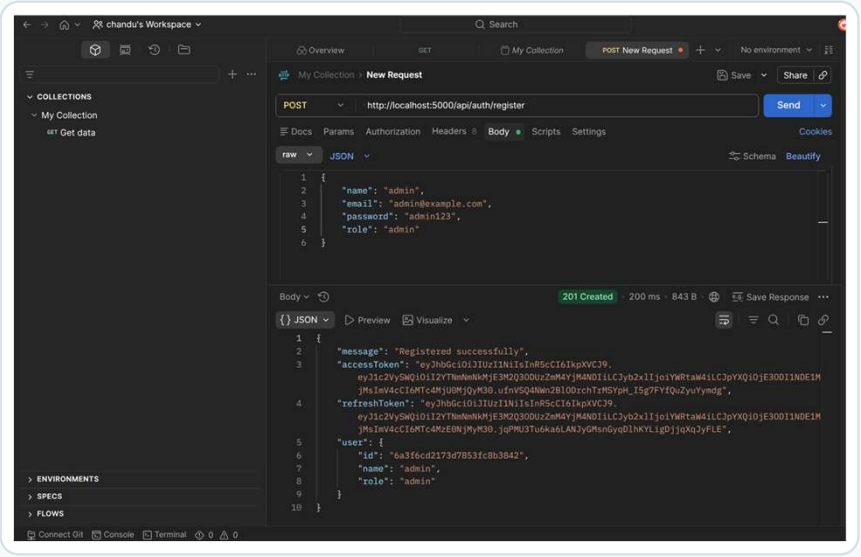


User schema (models/User.js)

Entry point, DB connection, auth, upload and Gemini modules are built as separate files.

API Testing with Postman

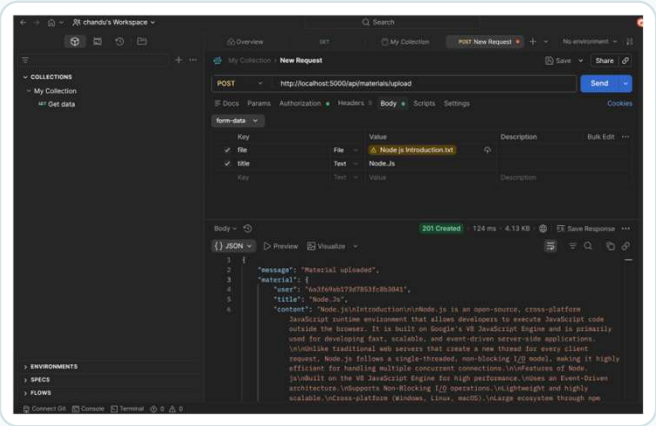
Endpoint	Purpose
POST /api/auth/register	Create user
POST /api/auth/login	Get JWT token
POST /api/materials/upload	Upload material
POST /api/materials/:id/summarize	AI summary
POST /api/ai/flashcards	AI flashcards
POST /api/ai/quiz	AI quiz
POST /api/ai/study-plan	AI study plan



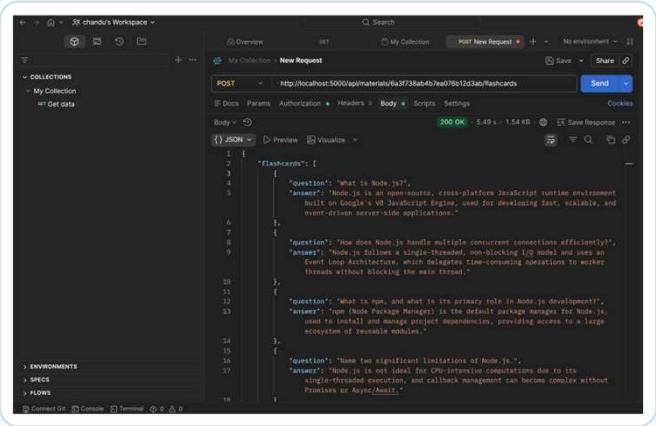
Registration request and response

All 7 core API tests passed.

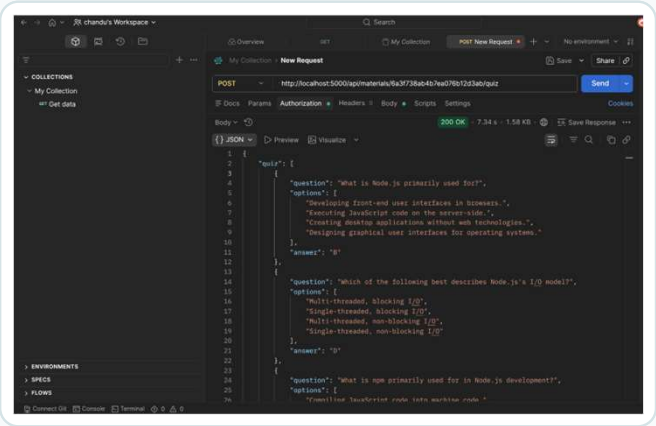
Demonstration Results



Material upload: 201 Created



AI flashcards



AI quiz

Rahul uploads his Node.js notes and gets a summary, flashcards, a quiz and a personalised study plan in minutes.

Manual preparation time drops; revision becomes focused and consistent.

Conclusion & Future Scope

- LearnMate delivers secure, scalable, AI-powered learning support.
- JWT, MongoDB and Gemini work together through a modular backend.
- Future: progress analytics, voice and handwritten-note input, more AI features.

Project links

Demo video

API testing video

Source code

Thank you